Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 50%. We keep your highest score.

Next item →

1. Which steps are involved in including the CIFAR-10 dataset in code for machine learning applications? — 1 / 1 point

- [ ] Resizing the images to 64x64 pixels
- [x] Normalizing the image data

  Correct
  Correct! Normalizing the image data is an important step to prepare the dataset for training.

- [ ] Converting the images to grayscale
- [x] Loading the dataset using a library-specific function

  Correct
  Correct! Loading the dataset using a function provided by a library like TensorFlow or PyTorch is essential.

- [x] Importing the required libraries

  Correct
  Correct! Importing the necessary libraries is an initial step in including the CIFAR-10 dataset in your code.

2. What is the first step in converting a random vector into a 32x32x3 output image using a GAN? — 1 / 1 point

- ( ) Reshaping the input vector to match the output image dimensions
- ( ) Defining the generator loss function
- (•) Initializing the random noise vector
- ( ) Training the discriminator network

  Correct
  Correct! Initializing the random noise vector is the first step in the process of generating an output image using a GAN.

3. Which of the following are types of downsampling techniques commonly used in discriminator models? — 1 / 1 point

- [x] Strided Convolution

  Correct
  Correct! Strided Convolution is also used for downsampling.

- [x] Average Pooling

  Correct
  Correct! Average Pooling is another common downsampling technique.

- [ ] Upsampling
- [x] Max Pooling

  Correct
  Correct! Max Pooling is a widely used downsampling technique.

- [ ] Padding

4. What is the primary role of the sigmoid activation function in a discriminator model? — 1 / 1 point

- ( ) To maintain the dimensionality of the input
- ( ) To perform element-wise multiplication
- ( ) To enhance the spatial resolution of the input
- (•) To map input values to a probability between 0 and 1

  Correct
  Correct! The sigmoid activation function maps input values to a probability between 0 and 1, which is essential for binary classification.

5. During the training of a GAN, which of the following issues can occur when the generator produces outputs that are too similar to the real data? — 1 / 1 point

- ( ) Overfitting
- ( ) Underfitting
- ( ) Vanishing gradient
- (•) Mode collapse

  Correct
  Correct! Mode collapse occurs when the generator produces outputs that are too similar, lacking diversity. Consider revisiting the section on mode collapse.